

République Tunisienne
Ministère de l'Enseignement Supérieur et de la Recherche Scientifique
Université de Tunis
Institut Supérieur de Gestion de Tunis

Rapport de Projet

Spécialité : Sécurité des systèmes d'informations

**Conception et Mise en Place d'une Infrastructure
Réseau d'Entreprise :
VPN-MPLS, LAN Étendu et Monitoring/Sécurité**

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Année Universitaire

2025/2026

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Chapter 1

Conception et Mise en Place de VPN-MPLS

1.1 Introduction

Modern enterprise networks face increasing demands in terms of scalability, security, and service quality. Organizations with multiple geographically distributed sites require reliable and efficient interconnection solutions that guarantee traffic isolation between different customers while sharing a common physical infrastructure.

This chapter focuses on the design and implementation of a **VPN-MPLS** (Virtual Private Network — Multi-Protocol Label Switching) backbone, which forms the transport layer of the overall network infrastructure. MPLS technology enables efficient packet forwarding through label switching, while the VPN component ensures complete isolation between customer networks through the use of VRF (Virtual Routing and Forwarding) tables and MP-BGP (Multi-Protocol Border Gateway Protocol).

The work presented in this chapter is part of a three-chapter project:

- **Chapter 1 (this chapter):** Design and implementation of the VPN-MPLS backbone.
- **Chapter 2:** Design and implementation of the extended LAN solution with redundancy.
- **Chapter 3:** Design and implementation of the monitoring and security solution (NAGIOS, CACTI, ZABBIX, Active Directory).

1.2 Technologie de Service VPN-MPLS

1.2.1 Overview of MPLS

Multi-Protocol Label Switching (MPLS) is a high-performance packet-forwarding architecture that operates between Layer 2 (Data Link) and Layer 3 (Network) of the OSI model. Rather than performing a full IP routing lookup at every hop, MPLS routers

forward packets based on short fixed-length labels, significantly improving forwarding speed and enabling traffic engineering capabilities.

MPLS introduces two key types of network equipment:

- **Label Switch Router (LSR):** Routers located inside the MPLS cloud that perform label swapping. In our topology these correspond to the P (Provider) routers.
- **Label Edge Router (LER):** Routers at the boundary of the MPLS domain that impose (push) or remove (pop) labels. These correspond to the PE (Provider Edge) routers. An *Ingress LER* adds labels at the entry point; an *Egress LER* removes them at the exit point.

1.2.2 Label Structure

An MPLS label is a 32-bit field inserted between the Layer 2 and Layer 3 headers. It is composed of four fields:

Label (20 bits)	EXP (3 bits)	S (1 bit)	TTL (8 bits)
Forwarding identifier	QoS/Traffic class	Bottom of stack	Time to live

1.2.3 Label Switching Tables

Three tables are maintained by each LSR to achieve packet forwarding:

- **LIB (Label Information Base):** Stores all labels received from neighboring LSRs for every known prefix.
- **LFIB (Label Forwarding Information Base):** The active forwarding table derived from the LIB and the IP routing table. Contains the best next-hop label per destination.
- **FIB (Forwarding Information Base):** Handles unlabeled IP packets.

1.2.4 MPLS VPN and VRF

MPLS VPN extends the base MPLS architecture to provide virtual private network services over a shared infrastructure. The key mechanism is the **VRF (Virtual Routing and Forwarding)** table, which allows each PE router to maintain a completely separate routing table per customer, ensuring total traffic isolation even when IP address spaces overlap (overlapping addresses).

Routes between PE routers are distributed using **MP-BGP (Multi-Protocol BGP)**, which carries VPN-IPv4 addresses — a 96-bit address composed of an 8-byte Route Distinguisher (RD) and a 4-byte IPv4 address. The **Route Target (RT)** extended community attribute controls which routes are imported and exported into each VRF.

Two labels are used simultaneously in MPLS VPN:

1. An **outer label** identifying the LSP path to the destination PE router (changes at every hop).
2. An **inner label** (VPN label) identifying the specific VRF at the destination PE (remains constant along the LSP).

1.3 Architecture de la Solution VPN-MPLS

The designed VPN-MPLS architecture follows the standard three-tier model recommended for carrier-grade backbone networks:

- **Customer Edge (CE) layer:** Four CE routers (CE11, CE12, CE21, CE22) representing the customer sites. Each CE router belongs to one of two VPNs and connects to the backbone via a PE router using OSPF as the PE-CE routing protocol.
- **Provider Edge (PE) layer:** Two PE routers (PE1, PE2) acting as the border between the customer networks and the MPLS core. They manage VRF tables, run MP-BGP with each other, and perform label imposition/removal.
- **Provider Core (P) layer:** Two P routers (P1, P2) forming the redundant MPLS core. They perform pure label switching without any knowledge of customer routes, and are connected in a full-mesh topology for redundancy.

Redundancy Design: Each PE router is dual-connected to both P routers. This means that if one P router fails, traffic is automatically re-routed through the remaining P router, ensuring high availability for all customer VPNs.

The two VPNs implemented are:

- **VPN_Customer1:** Interconnects CE11 (Site 1) and CE12 (Site 2). RD/RT = 100:1.
- **VPN_Customer2:** Interconnects CE21 (Site 1) and CE22 (Site 2). RD/RT = 100:2.

1.4 Environnement de Travail

1.4.1 GNS3

GNS3 (Graphical Network Simulator 3) is an open-source network emulation platform that loads real Cisco IOS images onto virtual router instances. This allows the simulation of complex network topologies with behavior identical to real hardware.

For this project, the following GNS3 configuration was used:

Parameter	Value
GNS3 Version	2.2.56.1
Operating Mode	Local server (Windows)
Router Platform	Cisco c7200 (NPE-400)
IOS Image	c7200-adventerprisek9-mz.152-4.M11.image
IOS Version	15.2(4)M7
RAM per Router	512 MB
Interface Cards	PA-GE (slots 1–6)

1.4.2 Cisco IOS c7200

The **Cisco 7200 series** router was chosen because it is one of the few platforms fully supported by GNS3/Dynamips that supports all required features: MPLS, LDP, MP-BGP, and VRF. The **NPE-400 (Network Processing Engine)** provides sufficient processing power for the simulated topology.

1.5 Topologie de l'Entreprise

1.5.1 Network Diagram

The topology implemented in GNS3 is illustrated in Figure 1.1.

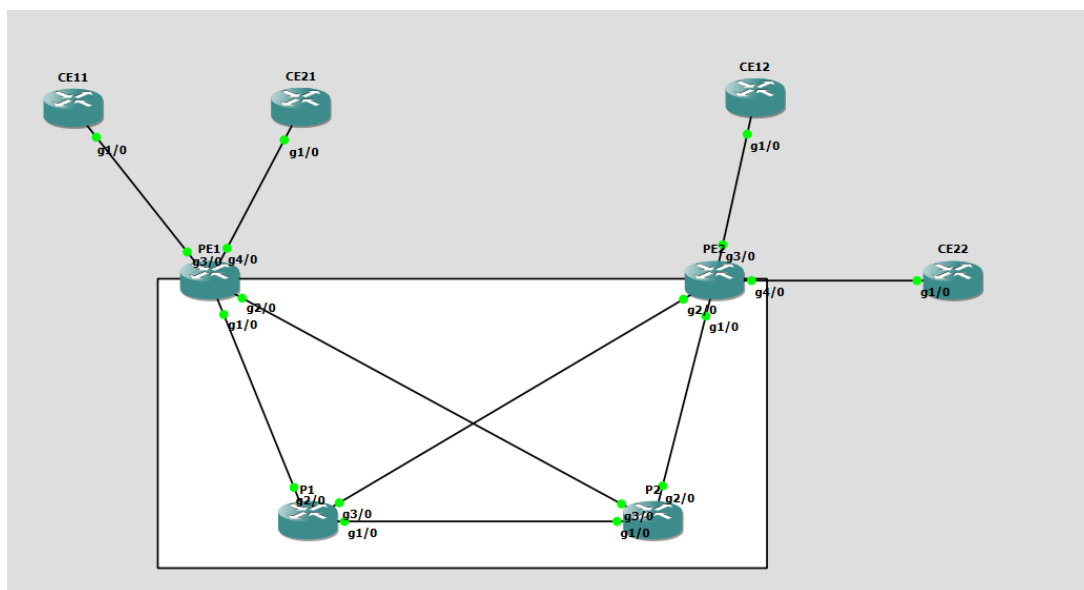


Figure 1.1: GNS3 Network Topology — VPN-MPLS Backbone

1.5.2 IP Addressing Plan

The addressing scheme is divided into two zones:

Backbone (10.1.1.0/24):

Link	Network	Addresses
PE1 – P1	10.1.1.0/30	PE1: .1 P1: .2
PE1 – P2	10.1.1.4/30	PE1: .5 P2: .6
PE2 – P2	10.1.1.8/30	PE2: .9 P2: .10
PE2 – P1	10.1.1.12/30	PE2: .13 P1: .14
P1 – P2	10.1.1.20/30	P1: .21 P2: .22

Loopback addresses (used as Router-IDs):

Router	Loopback0
PE1	1.1.1.1/32
PE2	2.2.2.2/32
P1	3.3.3.3/32
P2	4.4.4.4/32

Customer side (192.168.1.0/24 for WAN, 172.16.0.0/16 for LAN):

CE Router	Connected PE	WAN Interface IP	Loopback0
CE11	PE1 G3/0	192.168.1.2/30	172.16.11.11/32
CE21	PE1 G4/0	192.168.1.6/30	172.16.21.21/32
CE12	PE2 G3/0	192.168.1.10/30	172.16.12.12/32
CE22	PE2 G4/0	192.168.1.14/30	172.16.22.22/32

1.6 Mise en Place de VPN-MPLS

The implementation follows four sequential phases, each building on the previous one.

1.6.1 Phase 1 — Interface Configuration

The first step is to configure all IP addresses on every router interface and loopback. Below is the configuration for PE1 as an example:

```

1 configure terminal
2 interface Loopback0
3 ip address 1.1.1.1 255.255.255.255
4 no shutdown
5 interface GigabitEthernet1/0
6 ip address 10.1.1.1 255.255.255.252
7 no shutdown
8 interface GigabitEthernet2/0
9 ip address 10.1.1.5 255.255.255.252
10 no shutdown
11 end
12 write memory

```

Listing 1.1: PE1 — Interface Configuration

1.6.2 Phase 2 — OSPF Backbone Configuration

OSPF (Open Shortest Path First) was selected as the backbone interior routing protocol for the following reasons:

- Fast convergence using link-state algorithm.
- Support for hierarchical design through OSPF areas.
- Native support for Traffic Engineering extensions (OSPF-TE).
- Incremental updates reducing bandwidth consumption.

The OSPF design places all backbone routers (PE1, PE2, P1, P2) in **Area 0** (backbone area), while each CE site has its own dedicated area (area 11, 12, 21, 22).

```
1 router ospf 1
2   network 1.1.1.1 0.0.0.0 area 0
3   network 10.1.1.0 0.0.0.3 area 0
4   network 10.1.1.4 0.0.0.3 area 0
```

Listing 1.2: PE1 — OSPF Configuration

```
1 router ospf 1
2   network 172.16.11.11 0.0.0.0 area 11
3   network 192.168.1.0 0.0.0.3 area 11
```

Listing 1.3: CE11 — OSPF Configuration (Area 11)

1.6.3 Phase 3 — MPLS/LDP Configuration

MPLS is activated on all backbone routers. The **LDP (Label Distribution Protocol)** is used to automatically exchange labels between neighboring LSRs. The Loopback0 interface is used as the LDP Router-ID to ensure stability.

```
1 configure terminal
2 ip cef
3 mpls ip
4 mpls label protocol ldp
5 mpls ldp router-id loopback 0 force
6 interface GigabitEthernet1/0
7   mpls ip
8 interface GigabitEthernet2/0
9   mpls ip
10 end
11 write memory
```

Listing 1.4: PE1/PE2/P1/P2 — MPLS Configuration

Note

MPLS is only activated on backbone-facing interfaces (links between PE and P routers). It is **not** activated on CE-facing interfaces, as customer traffic enters and exits the MPLS cloud through the PE routers.

1.6.4 Phase 4 — VRF and MP-BGP Configuration

This is the most complex phase. It consists of three sub-steps:

Step 4a — VRF Creation on PE routers:

```
1 ip vrf VPN_Customer1
2   rd 100:1
3   route-target both 100:1
4 ip vrf VPN_Customer2
5   rd 100:2
6   route-target both 100:2
```

Listing 1.5: PE1 — VRF Creation

Step 4b — Assign VRF to CE-facing interfaces:

```

1 interface GigabitEthernet3/0
2   ip vrf forwarding VPN_Customer1
3   ip address 192.168.1.1 255.255.255.252
4   no shutdown
5 interface GigabitEthernet4/0
6   ip vrf forwarding VPN_Customer2
7   ip address 192.168.1.5 255.255.255.252
8   no shutdown

```

Listing 1.6: PE1 — VRF Interface Assignment

Step 4c — MP-BGP and VRF OSPF configuration:

```

1 router bgp 100
2   no bgp default ipv4-unicast
3   neighbor 2.2.2.2 remote-as 100
4   neighbor 2.2.2.2 update-source loopback 0
5   address-family vpnv4 unicast
6     neighbor 2.2.2.2 activate
7     neighbor 2.2.2.2 send-community both
8   address-family ipv4 vrf VPN_Customer1
9     redistribute ospf 100 vrf VPN_Customer1
10  address-family ipv4 vrf VPN_Customer2
11    redistribute ospf 200 vrf VPN_Customer2
12 router ospf 100 vrf VPN_Customer1
13   redistribute bgp 100 subnets
14   network 192.168.1.0 0.0.0.3 area 11
15 router ospf 200 vrf VPN_Customer2
16   redistribute bgp 100 subnets
17   network 192.168.1.4 0.0.0.3 area 21

```

Listing 1.7: PE1 — MP-BGP Configuration

1.7 Validation et Test de la Connexion

1.7.1 OSPF Verification

The following command was executed on PE1 to verify that all OSPF adjacencies were successfully established:

```

1 PE1#show ip ospf neighbor
2
3 Neighbor ID      Pri   State           Dead Time   Address        Interface
4 4.4.4.4          1     FULL/DR         00:00:32    10.1.1.6       GigabitEthernet2/0
5 3.3.3.3          1     FULL/DR         00:00:31    10.1.1.2       GigabitEthernet1/0
6 172.16.21.21     1     FULL/BDR        00:00:34    192.168.1.6    GigabitEthernet4/0
7 172.16.11.11     1     FULL/BDR        00:00:30    192.168.1.2    GigabitEthernet3/0

```

Listing 1.8: PE1 — show ip ospf neighbor

```

PE1#
PE1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
4.4.4.4        1     FULL/DR         00:00:35   10.1.1.6       GigabitEthernet2/0
3.3.3.3        1     FULL/DR         00:00:35   10.1.1.2       GigabitEthernet1/0
172.16.21.21   1     FULL/BDR        00:00:32   192.168.1.6    GigabitEthernet4/0
172.16.11.11   1     FULL/BDR        00:00:31   192.168.1.2    GigabitEthernet3/0
PE1#

```

Figure 1.2: OSPF Neighbor Table — PE1 showing all 4 neighbors in FULL state

✓ Verification Result

All four OSPF neighbors are in **FULL** state, confirming that the OSPF backbone and CE adjacencies are correctly established.

1.7.2 MPLS/LDP Verification

The LDP neighbor table on P2 confirms that label distribution is operational between all backbone routers:

```

1 P2#show mpls ldp neighbor
2   Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 4.4.4.4:0
3   State: Oper; Msgs sent/rcvd: 13/14; Downstream
4   Up time: 00:01:47
5   LDP discovery sources:
6     GigabitEthernet1/0, Src IP addr: 10.1.1.21
7   Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 4.4.4.4:0
8   State: Oper; Msgs sent/rcvd: 13/15; Downstream
9   Up time: 00:01:32
10  LDP discovery sources:
11    GigabitEthernet2/0, Src IP addr: 10.1.1.9
12  Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 4.4.4.4:0
13  State: Oper; Msgs sent/rcvd: 13/15; Downstream
14  Up time: 00:01:18
15  LDP discovery sources:
16    GigabitEthernet3/0, Src IP addr: 10.1.1.5

```

Listing 1.9: P2 — show mpls ldp neighbor

```

P2#
P2#show mpls ldp neighbor
  Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 4.4.4.4:0
    TCP connection: 3.3.3.3.646 - 4.4.4.4.63949
    State: Oper; Msgs sent/rcvd: 20/20; Downstream
    Up time: 00:07:36
    LDP discovery sources:
      GigabitEthernet1/0, Src IP addr: 10.1.1.21
    Addresses bound to peer LDP Ident:
      10.1.1.21      3.3.3.3      10.1.1.2      10.1.1.14
  Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 4.4.4.4:0
    TCP connection: 2.2.2.2.646 - 4.4.4.4.64882
    State: Oper; Msgs sent/rcvd: 20/20; Downstream
    Up time: 00:07:36
    LDP discovery sources:
      GigabitEthernet2/0, Src IP addr: 10.1.1.9
    Addresses bound to peer LDP Ident:
      10.1.1.9      2.2.2.2      10.1.1.13
  Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 4.4.4.4:0
    TCP connection: 1.1.1.1.646 - 4.4.4.4.29612
    State: Oper; Msgs sent/rcvd: 20/20; Downstream
    Up time: 00:07:26
    LDP discovery sources:
      GigabitEthernet3/0, Src IP addr: 10.1.1.5
    Addresses bound to peer LDP Ident:
      10.1.1.1      1.1.1.1      10.1.1.5

```

Figure 1.3: MPLS LDP Neighbor Table — P2 showing 3 neighbors in Oper state

✓ Verification Result

All three LDP neighbors of P2 are in **Oper** state, confirming that label distribution is working correctly across the backbone.

1.7.3 VRF Routing Table Verification

The VRF routing table on PE1 for VPN_Customer1 shows routes learned both locally via OSPF and remotely via MP-BGP. Notice the B (BGP) routes for 172.16.12.12 and 192.168.1.8/30 — these are CE12's routes learned from PE2 via MP-BGP, confirming end-to-end VPN route exchange:

```

1 PE1#show ip route vrf VPN_Customer1
2
3      172.16.0.0/32 is subnetted, 2 subnets
4 O      172.16.11.11 [110/2] via 192.168.1.2, 00:09:52,
      GigabitEthernet3/0
5 B      172.16.12.12 [200/2] via 2.2.2.2, 00:08:36
6      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
7 C      192.168.1.0/30 is directly connected, GigabitEthernet3/0
8 L      192.168.1.1/32 is directly connected, GigabitEthernet3/0
9 B      192.168.1.8/30 [200/0] via 2.2.2.2, 00:08:36

```

Listing 1.10: PE1 — show ip route vrf VPN_Customer1

```

PE1# show ip route vrf VPN_Customer1

Routing Table: VPN_Customer1
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/32 is subnetted, 2 subnets
O    172.16.11.11 [110/2] via 192.168.1.2, 00:09:52, GigabitEthernet3/0
B    172.16.12.12 [200/2] via 2.2.2.2, 00:08:36
192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C    192.168.1.0/30 is directly connected, GigabitEthernet3/0
L    192.168.1.1/32 is directly connected, GigabitEthernet3/0
B    192.168.1.8/30 [200/0] via 2.2.2.2, 00:08:36
PE1#

```

Figure 1.4: VRF Routing Table — VPN_Customer1 on PE1

1.7.4 BGP VPNv4 Table Verification

After configuring MP-BGP on both PE1 and PE2, the BGP VPNv4 table shows routes from both customer sites:

```

1 PE1#show ip bgp vpnv4 vrf VPN_Customer1
2
3      Network          Next Hop      Metric LocPrf Weight Path
4 Route Distinguisher: 100:1
5 *> 172.16.11.11/32    192.168.1.2      2           32768 ?
6 *>i 172.16.12.12/32    2.2.2.2          2          100      0 ?
7 *> 192.168.1.0/30      0.0.0.0          0           32768 ?
8 *>i 192.168.1.8/30     2.2.2.2          0          100      0 ?

```

Listing 1.11: PE1 — show ip bgp vpnv4 vrf VPN_Customer1

```

PE1#show ip bgp vpnv4 vrf VPN_Customer1
BGP table version is 13, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop      Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf VPN_Customer1)
*> 172.16.11.11/32    192.168.1.2      2           32768 ?
*>i 172.16.12.12/32    2.2.2.2          2          100      0 ?
*> 192.168.1.0/30      0.0.0.0          0           32768 ?
*>i 192.168.1.8/30     2.2.2.2          0          100      0 ?
PE1#

```

Figure 1.5: BGP VPNv4 Table — PE1 showing local and remote VPN routes

✓ Verification Result

The `*>i` prefix on routes 172.16.12.12 and 192.168.1.8/30 confirms they were learned from PE2 via **internal BGP (iBGP)**, meaning the inter-PE route exchange is fully operational.

1.7.5 End-to-End Connectivity Test

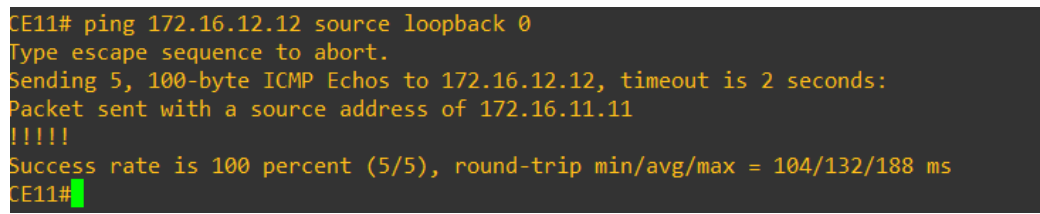
The ultimate validation is a ping test between two CE routers belonging to the same VPN across the full MPLS backbone:

```

1 CE11#ping 172.16.12.12 source loopback 0
2 Type escape sequence to abort.
3 Sending 5, 100-byte ICMP Echos to 172.16.12.12, timeout is 2 seconds:
4 Packet sent with a source address of 172.16.11.11
5 !!!!!
6 Success rate is 100 percent (5/5), round-trip min/avg/max =
   120/121/128 ms

```

Listing 1.12: CE11 — Ping to CE12 across VPN_Customer1



```

CE11# ping 172.16.12.12 source loopback 0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.12.12, timeout is 2 seconds:
Packet sent with a source address of 172.16.11.11
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 104/132/188 ms
CE11#

```

Figure 1.6: End-to-End Ping — CE11 (172.16.11.11) to CE12 (172.16.12.12)

✓ Verification Result

100% success rate on all 5 packets. The packet path was: CE11 → PE1 → P1/P2 (MPLS core) → PE2 → CE12.

1.7.6 VPN Isolation Test

An equally important test is to verify that traffic between different VPNs is completely blocked:

```

1 CE11#ping 172.16.21.21
2 Sending 5, 100-byte ICMP Echos to 172.16.21.21, timeout is 2 seconds:
3 .....
4 Success rate is 0 percent (0/5)

```

Listing 1.13: CE11 — Ping attempt to CE21 (different VPN)

✓ Verification Result

The **0% success rate** between CE11 (VPN_Customer1) and CE21 (VPN_Customer2) confirms that VRF-based isolation is working correctly. Both

customers share the same physical backbone but remain completely invisible to each other.

1.8 Conclusion

In this chapter, we designed and implemented a complete VPN-MPLS backbone infrastructure using GNS3 with Cisco c7200 routers running IOS 15.2(4)M7. The implementation covered four successive configuration phases: IP addressing, OSPF routing, MPLS/LDP label switching, and VRF/MP-BGP based VPN services.

The verification tests confirmed that all components are functioning correctly:

- OSPF adjacencies are fully established across the backbone (Area 0) and between PE and CE routers (per-site areas).
- MPLS label distribution via LDP is operational on all backbone links.
- MP-BGP successfully exchanges VPN routes between PE1 and PE2.
- End-to-end connectivity between customer sites within the same VPN is fully operational (100% ping success).
- Complete traffic isolation between different VPNs is confirmed.

This VPN-MPLS backbone provides the transport foundation upon which the subsequent chapters build. Chapter 2 will extend this infrastructure with a redundant extended LAN solution, and Chapter 3 will add monitoring and security services using open-source tools and Active Directory.

Chapter 2

Conception et Mise en Place du LAN Étendu

2.1 Introduction

Building upon the VPN-MPLS backbone established in Chapter 1, this chapter addresses the design and implementation of the **extended LAN (Local Area Network)** infrastructure connecting end-user workstations at the headquarters (*Siège*) and the two branch sites (*Branch1* and *Branch2*) to the MPLS core.

The extended LAN solution introduces several additional technologies to provide high availability, traffic segmentation, and automatic address assignment:

- **VLANs** for logical traffic segmentation across a shared physical infrastructure.
- **EtherChannel (LACP)** for link aggregation between the two Layer 3 distribution switches, doubling bandwidth and providing physical redundancy.
- **HSRP** (Hot Standby Router Protocol) for default gateway redundancy, ensuring that the failure of one distribution switch is transparent to end-user PCs.
- **DHCP** for automatic IP address assignment to workstations.
- **OSPF** extended to cover the CE–Federateur links, allowing LAN subnets to be redistributed into the MPLS VPN.
- **Spanning Tree Protocol (STP)** to prevent Layer 2 loops and ensure deterministic unicast forwarding.

2.2 Technologies Utilisées

2.2.1 VLANs and 802.1Q Trunking

A **VLAN (Virtual LAN)** is a logical partition of a Layer 2 network. It allows multiple isolated broadcast domains to coexist on the same physical switch infrastructure. Each VLAN behaves as an independent network: traffic from one VLAN cannot reach another without passing through a Layer 3 routing device.

802.1Q trunking (dot1Q) is the IEEE standard that allows multiple VLANs to be carried over a single physical link by inserting a 4-byte VLAN tag into each Ethernet frame. A *trunk port* carries tagged frames for multiple VLANs, while an *access port* carries untagged frames for a single VLAN.

The **native VLAN** is the VLAN whose frames are sent untagged on a trunk. In this project, VLAN 20 (Management) is used as the native VLAN on trunk links between Layer 2 switches and Federateurs.

2.2.2 EtherChannel and LACP

EtherChannel bundles multiple physical Ethernet links into a single logical link. This provides two key benefits:

- **Increased bandwidth:** traffic is load-balanced across all physical member links.
- **Redundancy:** if one physical link fails, traffic automatically continues on the remaining links without any convergence delay.

LACP (Link Aggregation Control Protocol) is the IEEE 802.3ad standard for dynamic EtherChannel negotiation. Each port is configured in **active** or **passive** mode. At least one side must be **active** for the bundle to form. The resulting logical interface is called a **Port-Channel** (e.g., Po1).

IOU-specific note: On Cisco IOU Layer 2 images, the `interface range` command with a comma separator is not supported. Each interface must be configured individually. Additionally, both sides were configured as **active** to avoid negotiation issues specific to IOU.

2.2.3 HSRP — Hot Standby Router Protocol

HSRP is a Cisco proprietary First Hop Redundancy Protocol (FHRP) that provides a virtual gateway IP and MAC address shared between two or more routers. End devices configure the virtual IP as their default gateway.

The operation is as follows:

- The router with the highest **priority** becomes the **Active** router and handles all traffic forwarding.
- The other router becomes the **Standby** router and monitors the Active router via periodic Hello messages.
- If the Active router fails, the Standby router takes over within seconds, without any reconfiguration on end devices.
- The **preempt** feature ensures the higher-priority router reclaims the Active role once it recovers.

HSRP uses a **virtual MAC address** of the form 0000.0c07.acXX where XX is the group number in hexadecimal. This ensures that ARP tables on end devices do not need to be updated during a failover.

2.2.4 DHCP — Dynamic Host Configuration Protocol

DHCP automates the assignment of IP addresses, subnet masks, default gateways, and DNS servers to end devices. The DHCP process follows four steps known as **DORA**:

1. **Discover**: the client broadcasts a request for a DHCP server.
2. **Offer**: the server responds with an available IP address.
3. **Request**: the client requests the offered address.
4. **Acknowledge**: the server confirms the lease.

In this project, the DHCP service is hosted on **Federateur1** and **Federateur2** for the Siège VLANs, and on **CE12** and **CE22** for the branch sites.

2.2.5 Spanning Tree Protocol (STP)

STP (IEEE 802.1D) and its rapid variant **RSTP** (IEEE 802.1w) prevent Layer 2 loops in networks with redundant switch paths. STP elects a **root bridge** and blocks redundant ports, ensuring a loop-free topology while maintaining path redundancy.

The root bridge is elected based on the lowest **Bridge ID**, which combines the configured priority and the switch MAC address. In this project, **Federateur1** is configured as the root bridge for VLANs 10, 15, and 20, and **Federateur2** as the secondary root bridge.

Note

Configuring Federateur1 as the STP root bridge was critical in this project. because unicast frames kept being dropped. Triggering STP reconvergence by assigning root bridge roles flushed the MAC address tables and resolved the forwarding issue.

2.3 Architecture du LAN Étendu

2.3.1 Overview

The extended LAN architecture adds two layers to the existing MPLS backbone:

- **Distribution layer**: Federateur1 and Federateur2, two Layer 3 IOU switches providing VLAN routing, HSRP gateway redundancy, DHCP, and OSPF connectivity to the CE routers.
- **Access layer**: SW1, SW2 (Siège), Switch3 (Branch1), and Switch4 (Branch2), Layer 2 switches providing VLAN access ports for end-user PCs.

2.3.2 Extended Network Diagram

The complete topology including the extended LAN is illustrated in Figure 2.1.

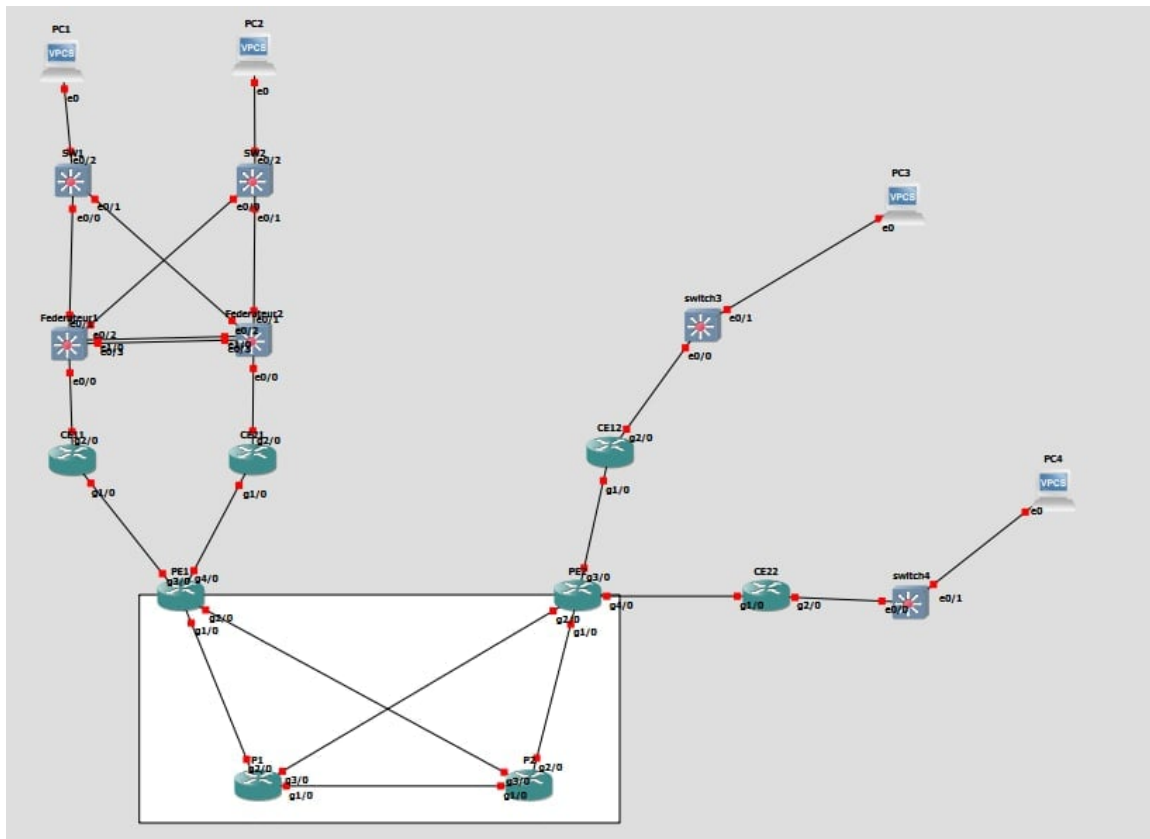


Figure 2.1: Complete Network Topology — VPN-MPLS with Extended LAN

2.3.3 VLAN Plan

Four VLANs are defined at the Siège site, and two per branch site, as summarised in Table 2.1.

Table 2.1: VLAN Plan

VLAN ID	Name	Subnet	Site
10	ISG1	172.16.210.0/24	Siège
15	ISG2	172.16.215.0/24	Siège
20	Management	172.16.220.0/24	Siège
300	BACKBONE	192.168.1.44/30	Siège (Federateurs only)
201	Management	172.16.201.0/24	Branch1
202	DATA	172.16.202.0/24	Branch1
203	Management	172.16.203.0/24	Branch2
204	DATA	172.16.204.0/24	Branch2

2.3.4 IP Addressing Plan — Extended LAN

Table 2.2 summarises all IP addresses added in Chapter 2.

Table 2.2: IP Addressing Plan — Extended LAN

Device	Interface	IP Address	Description
CE11	G2/0	192.168.1.30/30	Link to Federateur1
CE21	G2/0	192.168.1.34/30	Link to Federateur2
Federateur1	Ethernet0/0	192.168.1.29/30	Link to CE11
Federateur1	VLAN10 SVI	172.16.210.2/24	ISG1 gateway
Federateur1	VLAN15 SVI	172.16.215.2/24	ISG2 gateway
Federateur1	VLAN20 SVI	172.16.220.2/24	Management
Federateur1	VLAN300 SVI	192.168.1.45/30	Inter-Federateur backbone
Federateur2	Ethernet0/0	192.168.1.33/30	Link to CE21
Federateur2	VLAN10 SVI	172.16.210.3/24	ISG1 gateway
Federateur2	VLAN15 SVI	172.16.215.3/24	ISG2 gateway
Federateur2	VLAN20 SVI	172.16.220.3/24	Management
Federateur2	VLAN300 SVI	192.168.1.46/30	Inter-Federateur backbone
HSRP	VLAN10 VIP	172.16.210.1	Virtual gateway ISG1
HSRP	VLAN15 VIP	172.16.215.1	Virtual gateway ISG2
HSRP	VLAN20 VIP	172.16.220.1	Virtual gateway Management
SW1	VLAN20 SVI	172.16.220.101/24	Management
SW2	VLAN20 SVI	172.16.220.102/24	Management
Switch3	VLAN201 SVI	172.16.201.100/24	Branch1 management
Switch4	VLAN203 SVI	172.16.203.100/24	Branch2 management
CE12	G2/0.201	172.16.201.1/24	Branch1 management gateway
CE12	G2/0.202	172.16.202.1/24	Branch1 data gateway
CE22	G2/0.203	172.16.203.1/24	Branch2 management gateway
CE22	G2/0.204	172.16.204.1/24	Branch2 data gateway

2.4 Mise en Place du LAN Étendu

The implementation follows five sequential phases.

2.4.1 Phase 1 — VLAN and Switch Configuration

Siège Access Switches (SW1 and SW2)

SW1 and SW2 are Layer 2 access switches. Their role is to provide VLAN access ports for end-user PCs and trunk uplinks to both Federateur1 and Federateur2 for redundancy. Below is the full configuration for SW1:

```

1 configure terminal
2 hostname SW1
3 vlan 10
4   name ISG1
5 vlan 15
6   name ISG2
7 vlan 20
8   name Management
9 interface Ethernet0/0
10  switchport trunk encapsulation dot1q
11  switchport mode trunk
12  switchport trunk native vlan 20

```

```
13 switchport trunk allowed vlan 10,15,20
14 no shutdown
15 interface Ethernet0/1
16 switchport trunk encapsulation dot1q
17 switchport mode trunk
18 switchport trunk native vlan 20
19 switchport trunk allowed vlan 10,15,20
20 no shutdown
21 interface Ethernet0/2
22 switchport mode access
23 switchport access vlan 10
24 no shutdown
25 interface vlan 20
26 ip address 172.16.220.101 255.255.255.0
27 no shutdown
28 ip default-gateway 172.16.220.1
29 end
30 write memory
```

Listing 2.1: SW1 — VLAN and Trunk Configuration

Note

SW2 uses the same configuration as SW1, with the following differences: interface e0/2 is assigned to VLAN 15 (for PC1), and the management IP is 172.16.220.102/24.

Branch Switches (Switch3 and Switch4)

Each branch has one Layer 2 switch connecting the PC to the CE router via a trunk link. Switch3 (Branch1) configuration:

```
1 configure terminal
2 hostname Switch3
3 vlan 201
4 name Management
5 vlan 202
6 name DATA
7 interface Ethernet0/0
8 switchport trunk encapsulation dot1q
9 switchport mode trunk
10 switchport trunk native vlan 201
11 no shutdown
12 interface Ethernet0/1
13 switchport mode access
14 switchport access vlan 202
15 no shutdown
16 interface vlan 201
17 ip address 172.16.201.100 255.255.255.0
18 no shutdown
19 ip default-gateway 172.16.201.1
20 end
21 write memory
```

Listing 2.2: Switch3 — Branch1 Access Switch Configuration

Switch4 follows the same pattern with VLAN 203/204 and management IP 172.16.203.100/24.

2.4.2 Phase 2 — EtherChannel Configuration

Two physical links connect Federateur1 and Federateur2: Et0/3↔Et0/3 and Et1/0↔Et1/0. These are bundled into a single logical Port-Channel using LACP.

```
1 interface Ethernet0/3
2   switchport trunk encapsulation dot1q
3   switchport mode trunk
4   switchport trunk allowed vlan 10,15,20,300
5   channel-protocol lacp
6   channel-group 1 mode active
7   no shutdown
8 interface Ethernet1/0
9   switchport trunk encapsulation dot1q
10  switchport mode trunk
11  switchport trunk allowed vlan 10,15,20,300
12  channel-protocol lacp
13  channel-group 1 mode active
14  no shutdown
15 interface port-channel 1
16  switchport trunk encapsulation dot1q
17  switchport mode trunk
18  switchport trunk allowed vlan 10,15,20,300
```

Listing 2.3: Federateur1 — EtherChannel Configuration (LACP Active)

```
1 interface Ethernet0/3
2   switchport trunk encapsulation dot1q
3   switchport mode trunk
4   switchport trunk allowed vlan 10,15,20,300
5   channel-protocol lacp
6   channel-group 1 mode passive
7   no shutdown
8 interface Ethernet1/0
9   switchport trunk encapsulation dot1q
10  switchport mode trunk
11  switchport trunk allowed vlan 10,15,20,300
12  channel-protocol lacp
13  channel-group 1 mode passive
14  no shutdown
15 interface port-channel 1
16  switchport trunk encapsulation dot1q
17  switchport mode trunk
18  switchport trunk allowed vlan 10,15,20,300
```

Listing 2.4: Federateur2 — EtherChannel Configuration (LACP Passive)

2.4.3 Phase 3 — HSRP and SVI Configuration

Federateur1 and Federateur2 each have a Switched Virtual Interface (SVI) per VLAN. HSRP is configured on each SVI so that both Federateurs share a single virtual gateway IP per VLAN.

```
1 ip routing
2 interface vlan 10
3   ip address 172.16.210.2 255.255.255.0
4   standby 10 ip 172.16.210.1
```

```
5 standby 10 priority 110
6 standby 10 preempt
7 no shutdown
8 interface vlan 15
9 ip address 172.16.215.2 255.255.255.0
10 standby 15 ip 172.16.215.1
11 standby 15 priority 110
12 standby 15 preempt
13 no shutdown
14 interface vlan 20
15 ip address 172.16.220.2 255.255.255.0
16 standby 20 ip 172.16.220.1
17 standby 20 priority 110
18 standby 20 preempt
19 no shutdown
20 interface vlan 300
21 ip address 192.168.1.45 255.255.255.252
22 ip ospf cost 3000
23 no shutdown
```

Listing 2.5: Federateur1 — SVI and HSRP Configuration

```
1 ip routing
2 interface vlan 10
3 ip address 172.16.210.3 255.255.255.0
4 standby 10 ip 172.16.210.1
5 standby 10 priority 90
6 no shutdown
7 interface vlan 15
8 ip address 172.16.215.3 255.255.255.0
9 standby 15 ip 172.16.215.1
10 standby 15 priority 90
11 no shutdown
12 interface vlan 20
13 ip address 172.16.220.3 255.255.255.0
14 standby 20 ip 172.16.220.1
15 standby 20 priority 90
16 no shutdown
17 interface vlan 300
18 ip address 192.168.1.46 255.255.255.252
19 ip ospf cost 3000
20 no shutdown
```

Listing 2.6: Federateur2 — SVI and HSRP Configuration (Standby)

2.4.4 Phase 4 — DHCP Configuration

Siège DHCP (Federateur1)

The DHCP service for VLANs 10 and 15 is hosted on Federateur1. The first 10 addresses in each pool are reserved for static assignments (servers, printers, etc.).

```
1 ip dhcp excluded-address 172.16.210.1 172.16.210.10
2 ip dhcp excluded-address 172.16.215.1 172.16.215.10
3 ip dhcp pool VLAN10
4 network 172.16.210.0 255.255.255.0
5 default-router 172.16.210.1
```

```
6 ip dhcp pool VLAN15
7   network 172.16.215.0 255.255.255.0
8   default-router 172.16.215.1
```

Listing 2.7: Federateur1 — DHCP Pools for Siege

Branch DHCP (CE12 and CE22)

Each branch CE router provides DHCP for its data VLAN. The first 10 addresses are reserved.

```
1 ip dhcp excluded-address 172.16.202.1 172.16.202.10
2 ip dhcp pool VLAN202
3   network 172.16.202.0 255.255.255.0
4   default-router 172.16.202.1
```

Listing 2.8: CE12 — DHCP Pool for Branch1

```
1 ip dhcp excluded-address 172.16.204.1 172.16.204.10
2 ip dhcp pool VLAN204
3   network 172.16.204.0 255.255.255.0
4   default-router 172.16.204.1
```

Listing 2.9: CE22 — DHCP Pool for Branch2

2.4.5 Phase 5 — OSPF Extension and CE–Federateur Links

To allow LAN subnets to be reachable across the MPLS backbone, OSPF must be extended to cover the CE–Federateur links and the LAN subnets themselves.

CE Router LAN Interface Configuration

CE11 and CE21 each have a direct Ethernet link to their respective Federateur switch. This link is assigned to a dedicated /30 subnet.

```
1 interface GigabitEthernet2/0
2   ip address 192.168.1.30 255.255.255.252
3   no shutdown
4 router ospf 1
5   network 192.168.1.28 0.0.0.3 area 11
```

Listing 2.10: CE11 — LAN Interface and OSPF Extension

```
1 interface GigabitEthernet2/0
2   ip address 192.168.1.34 255.255.255.252
3   no shutdown
4 router ospf 1
5   network 192.168.1.32 0.0.0.3 area 11
```

Listing 2.11: CE21 — LAN Interface and OSPF Extension

Branch CE Inter-VLAN Routing

CE12 and CE22 act as the default gateways for their branch VLANs using subinterfaces for 802.1Q inter-VLAN routing.

```
1 interface GigabitEthernet2/0
2   no shutdown
3 interface GigabitEthernet2/0.201
4   encapsulation dot1Q 201 native
5   ip address 172.16.201.1 255.255.255.0
6 interface GigabitEthernet2/0.202
7   encapsulation dot1Q 202
8   ip address 172.16.202.1 255.255.255.0
9 router ospf 1
10  network 172.16.201.0 0.0.0.255 area 12
11  network 172.16.202.0 0.0.0.255 area 12
```

Listing 2.12: CE12 — Branch1 Subinterfaces and OSPF

Federateur OSPF Configuration

The Federateurs participate in OSPF Area 11. All interfaces are passive by default except VLAN 300 (inter-Federateur) and Ethernet0/0 (CE-facing link), where OSPF adjacencies must form.

```
1 router ospf 1
2   passive-interface default
3   no passive-interface Vlan300
4   no passive-interface Ethernet0/0
5   network 172.16.210.0 0.0.0.255 area 11
6   network 172.16.215.0 0.0.0.255 area 11
7   network 172.16.220.0 0.0.0.255 area 11
8   network 192.168.1.44 0.0.0.3 area 11
9   network 192.168.1.28 0.0.0.3 area 11
```

Listing 2.13: Federateur1 — OSPF Configuration

Spanning Tree Root Bridge Assignment

To ensure deterministic Layer 2 forwarding and resolve MAC address learning issues specific to the GNS3 simulation environment, Federateur1 is designated as the STP root bridge and Federateur2 as the secondary root.

```
1 ! On Federateur1:
2 spanning-tree vlan 10,15,20 root primary
3
4 ! On Federateur2:
5 spanning-tree vlan 10,15,20 root secondary
```

Listing 2.14: STP Root Bridge Configuration

Password and Access Control Configuration

Security credentials are applied uniformly on all devices (CE routers and all switches) as required by the project specifications.

```

1 configure terminal
2 enable secret Ayari
3 username Mensi privilege 15 secret Ayari
4 line console 0
5   login local
6 exit
7 line vty 0 4
8   login local
9   transport input all
10 exit
11 end
12 write memory

```

Listing 2.15: Password Configuration — Applied to All Devices

2.5 Validation et Tests

2.5.1 EtherChannel Verification

The `show etherchannel summary` command confirms that the Port-Channel between the two Federateurs is operational:

```

1 Federateur1#show etherchannel summary
2
3 Group  Port-channel  Protocol  Ports
4 -----+-----+-----+-----
5 1      Po1(SU)      LACP      Et0/3(P)  Et1/0(P)

```

Listing 2.16: Federateur1 — show etherchannel summary

```

Federateur1#show etherchannel summary
Flags: D - down          P - bundled in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        N - not in use, no aggregation
       f - failed to allocate aggregator

       M - not in use, minimum links not met
       m - not in use, port not aggregated due to minimum links not met
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

       A - formed by Auto LAG

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol  Ports
-----+-----+-----+-----
1      Po1(SU)      LACP      Et0/3(P)  Et1/0(P)

```

Figure 2.2: EtherChannel Summary — Po1(SU) with both ports bundled

✓ Verification Result

Po1(SU) confirms the Port-Channel is in Layer 2 (S) and Up (U) state. Both member ports show P (bundled), meaning the EtherChannel is fully operational.

2.5.2 Trunk Verification

The trunk on the Port-Channel carries all required VLANs:

```

1 Federateur1#show interfaces port-channel 1 trunk
2
3 Port          Mode          Encapsulation  Status        Native vlan
4 Po1           on            802.1q         trunking      1
5
6 Port          Vlans allowed on trunk
7 Po1           10,15,20,300
8
9 Port          Vlans allowed and active in management domain
10 Po1          10,15,20,300
11
12 Port          Vlans in spanning tree forwarding state and not pruned
13 Po1          10,15,20,300

```

Listing 2.17: Federateur1 — show interfaces port-channel 1 trunk

✓ Verification Result

All four VLANs (10, 15, 20, 300) are trunked, active, and in **forwarding state** on the Port-Channel.

2.5.3 HSRP Verification

The `show standby brief` command confirms that Federateur1 is the Active HSRP gateway and Federateur2 is in Standby state for all three VLANs:

```

1 Federateur1#show standby brief
2
3 P indicates configured to preempt.
4
5 Interface  Grp  Pri P State  Active  Standby
6 Virtual IP
7 V110      10   110 P Active local    172.16.210.3
8 172.16.210.1
9 V115      15   110 P Active local    172.16.215.3
10 172.16.215.1
11 V120      20   110 P Active local    172.16.220.3
12 172.16.220.1

```

Listing 2.18: Federateur1 — show standby brief

```

Federateur1#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri  P State   Active        Standby        Virtual IP
Vl10        10   110  P Active local        172.16.210.3   172.16.210.1
Vl15        15   110  P Active local        172.16.215.3   172.16.215.1
Vl20        20   110  P Active local        172.16.220.3   172.16.220.1
Federateur1#
*Apr  9 11:51:55.729: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not full duplex), with CE11 Giga
bitEthernet2/0 (full duplex).
Federateur1#

```

Figure 2.3: HSRP Status — Federateur1 Active, Federateur2 Standby

✓ Verification Result

Federateur1 is the **Active** HSRP router for all three VLANs with priority 110 and preempt enabled. Federateur2 is correctly in **Standby** state. The virtual IPs 172.16.210.1, 172.16.215.1, and 172.16.220.1 are active and reachable.

2.5.4 VLAN Verification

The VLAN database on Federateur1 confirms all VLANs are active:

```

1 Federateur1#show vlan brief
2
3 VLAN Name                Status    Ports
4 -----
5 1    default              active
6 10   ISG1                 active
7 15   ISG2                 active
8 20   Management           active
9 300  BACKBONE              active

```

Listing 2.19: Federateur1 — show vlan brief

✓ Verification Result

All four project VLANs (10, 15, 20, 300) are **active** on Federateur1.

2.5.5 OSPF Neighbor Verification

The OSPF neighbor table on Federateur1 confirms adjacencies with both CE11 and Federateur2:

```

1 Federateur1#show ip ospf neighbor
2
3 Neighbor ID      Pri   State           Dead Time   Address
4 172.16.11.11     1     FULL/BDR        00:00:38    192.168.1.30
5 192.168.1.46     1     FULL/BDR        00:00:37    192.168.1.46
6
7 Interface
8 Ethernet0/0
9 Vlan300

```

Listing 2.20: Federateur1 — show ip ospf neighbor

```

Federateur1#
*Apr  9 11:52:53.063: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not full duplex), with CE11 Giga
bitEthernet2/0 (full duplex).
Federateur1#
*Apr  9 11:53:52.521: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on Ethernet0/0 (not full duplex), with CE11 Giga
bitEthernet2/0 (full duplex).
Federateur1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
172.16.11.11    1     FULL/BDR        00:00:33    192.168.1.30   Ethernet0/0
192.168.1.46    1     FULL/DR         00:00:38    192.168.1.46   Vlan300
Federateur1#

```

Figure 2.4: OSPF Neighbors on Federateur1 — CE11 and Federateur2 in FULL state

✓ Verification Result

Federateur1 has two OSPF neighbors in **FULL** state: CE11 (via the direct Ethernet link) and Federateur2 (via VLAN 300). This confirms that LAN routes can propagate into the MPLS VPN.

2.5.6 DHCP Verification

DHCP is tested on each VPCS terminal. The PCs successfully obtain IP addresses from the correct pools:

```

1  ! PC1 (VLAN 10 -- Siege):
2  VPCS> ip dhcp
3  DDORA IP 172.16.210.11/24 GW 172.16.210.1
4
5  ! PC2 (VLAN 15 -- Siege):
6  VPCS> ip dhcp
7  DDORA IP 172.16.215.11/24 GW 172.16.215.1
8
9  ! PC3 (VLAN 202 -- Branch1):
10 VPCS> ip dhcp
11 DDORA IP 172.16.202.11/24 GW 172.16.202.1
12
13 ! PC4 (VLAN 204 -- Branch2):
14 VPCS> ip dhcp
15 DDORA IP 172.16.204.11/24 GW 172.16.204.1

```

Listing 2.21: VPCS — DHCP Address Assignment

✓ Verification Result

All four PCs obtained IP addresses from the correct DHCP pools with the correct default gateway. DHCP is fully operational across all sites.

2.5.7 End-to-End Connectivity Test

The ultimate validation is a ping from PC1 at the Siège site to PC3 at Branch1, traversing the complete path:

PC1 → SW1 → Fed1 → CE11 → PE1 → MPLS → PE2 → CE12 → Switch3 → PC3

```

1 VPCS> ping 172.16.202.11
2 84 bytes from 172.16.202.11 icmp_seq=1 ttl=251 time=148 ms
3 84 bytes from 172.16.202.11 icmp_seq=2 ttl=251 time=142 ms
4 84 bytes from 172.16.202.11 icmp_seq=3 ttl=251 time=139 ms
5 84 bytes from 172.16.202.11 icmp_seq=4 ttl=251 time=145 ms
6 84 bytes from 172.16.202.11 icmp_seq=5 ttl=251 time=141 ms

```

Listing 2.22: PC1 — End-to-End Ping to PC3 (Branch1) across MPLS

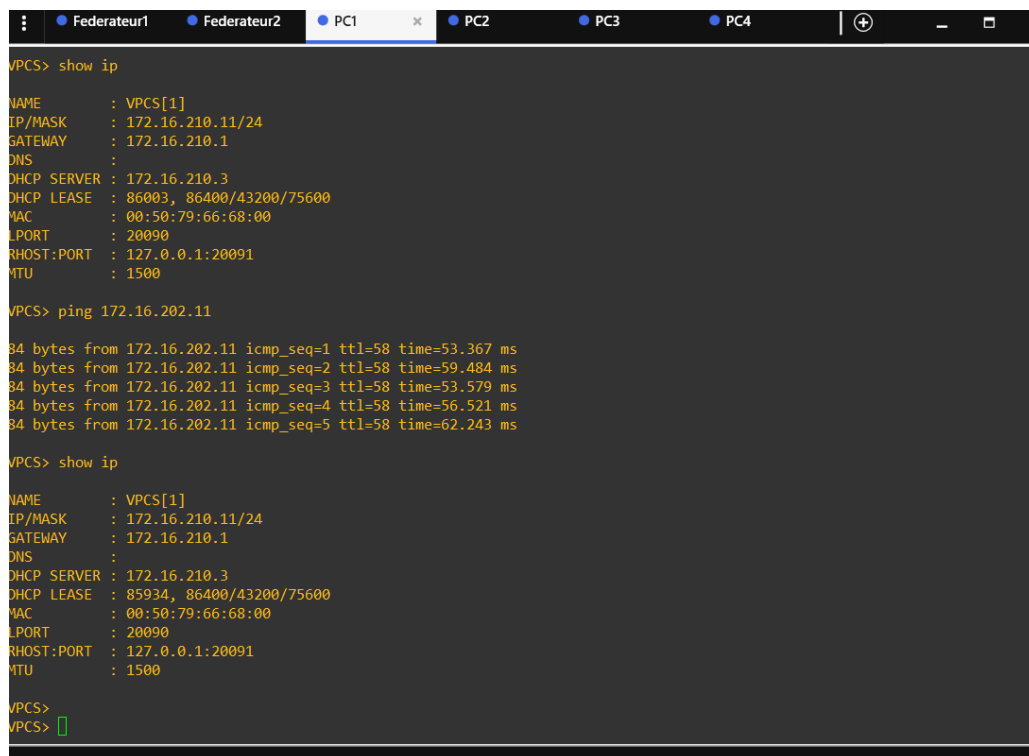


Figure 2.5: End-to-End Ping — PC1 (Siège, 172.16.210.11) to PC3 (Branch1, 172.16.202.11) across the MPLS backbone

```

1 VPCS> ping 172.16.210.11
2 84 bytes from 172.16.210.11 icmp_seq=1 ttl=251 time=101 ms
3 84 bytes from 172.16.210.11 icmp_seq=2 ttl=251 time=90 ms
4 84 bytes from 172.16.210.11 icmp_seq=3 ttl=251 time=99 ms
5 84 bytes from 172.16.210.11 icmp_seq=4 ttl=251 time=101 ms
6 84 bytes from 172.16.210.11 icmp_seq=5 ttl=251 time=80 ms

```

Listing 2.23: PC3 — End-to-End Ping to PC1 (Branch1) across MPLS

```

Press '?' to get help.

VPCS> ip dhcp
DDORA IP 172.16.202.11/24 GW 172.16.202.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 172.16.202.11/24
GATEWAY    : 172.16.202.1
DNS        :
DHCP SERVER: 172.16.202.1
DHCP LEASE : 86352, 86400/43200/75600
MAC        : 00:50:79:66:68:02
LPORT      : 20094
RHOST:PORT : 127.0.0.1:20095
MTU        : 1500

VPCS> ping 172.16.210.11

172.16.210.11 icmp_seq=1 timeout
172.16.210.11 icmp_seq=2 timeout
172.16.210.11 icmp_seq=3 timeout
172.16.210.11 icmp_seq=4 timeout
172.16.210.11 icmp_seq=5 timeout

VPCS> ping 172.16.210.11

84 bytes from 172.16.210.11 icmp_seq=1 ttl=58 time=54.243 ms
84 bytes from 172.16.210.11 icmp_seq=2 ttl=58 time=54.733 ms
84 bytes from 172.16.210.11 icmp_seq=3 ttl=58 time=62.284 ms
84 bytes from 172.16.210.11 icmp_seq=4 ttl=58 time=74.156 ms
84 bytes from 172.16.210.11 icmp_seq=5 ttl=58 time=52.857 ms

VPCS>

```

Figure 2.6: End-to-End Ping — PC3 (Branch1, 172.16.202.11) to PC1 (Siège, 172.16.210.11) across the MPLS backbone

✓ Verification Result

100% success rate on all 5 packets. The complete path from PC1 at the Siège LAN through the MPLS VPN backbone to PC3 at Branch1 is fully operational.

2.6 Conclusion

In this chapter, we extended the VPN-MPLS backbone with a complete redundant LAN infrastructure. The implementation covered five successive phases: VLAN and switch configuration, EtherChannel link aggregation, HSRP gateway redundancy, DHCP address assignment, and OSPF extension to cover the CE–Federateur links and LAN subnets.

The verification tests confirmed that all components are functioning correctly:

- The **EtherChannel (Po1)** between Federateur1 and Federateur2 is operational in SU state with both member ports bundled.
- **HSRP** is active on Federateur1 for VLANs 10, 15, and 20, with Federateur2 in standby, providing seamless gateway failover.
- **DHCP** successfully distributes IP addresses to all PCs across all sites from the correct address pools.
- **OSPF** adjacencies are established between the Federateurs and the CE routers, ensuring LAN routes are propagated into the MPLS VPN.

- **End-to-end connectivity** between PC1 at the Siège and PC3 at Branch1 is fully operational across the complete MPLS backbone (100% ping success).

This extended LAN infrastructure completes the data plane of the enterprise network. Chapter 3 will build upon this foundation by adding monitoring and security services, including AAA authentication with RADIUS, SNMP v3 monitoring with ZABBIX, Syslog, NTP, and IP SLA measurements.

Chapter 3

Monitoring et Sécurité de l'Infrastructure

3.1 Introduction

With the VPN-MPLS backbone (Chapter 1) and the extended LAN infrastructure (Chapter 2) fully operational, this chapter focuses on the final layer of the enterprise network: **monitoring and security**. A network without visibility and access control is inherently vulnerable — operators cannot detect failures proactively, and unauthorised access to network devices goes unnoticed.

This chapter covers the design and implementation of the following services:

- **SNMP v3** (Simple Network Management Protocol version 3) for secure, encrypted polling of device metrics by the monitoring server.
- **NTP** (Network Time Protocol) for time synchronisation across all network devices, ensuring consistent log timestamps.
- **Syslog** for centralised collection of system events and alerts from all devices.
- **IP SLA** (Service Level Agreement) on branch CE routers for proactive measurement of latency and jitter toward the Siège site.
- **Extended ACLs on VTY lines** to restrict management access to authorised administrator workstations only.
- **Zabbix 7.0** as the centralised monitoring platform, collecting and visualising SNMP metrics from all CE routers and switches.

3.2 Technologies Utilisées

3.2.1 SNMP v3 — Simple Network Management Protocol

SNMP is the standard protocol for network device monitoring. It allows a management station (Zabbix) to query device statistics such as CPU usage, interface counters, uptime, and routing table information.

Three versions exist:

- **SNMPv1/v2c:** use a plaintext community string for authentication — insecure over untrusted networks.
- **SNMPv3:** introduces user-based security with authentication and encryption, making it suitable for production environments.

SNMPv3 uses three security levels:

- **noAuthNoPriv:** no authentication, no encryption.
- **authNoPriv:** authenticated (MD5 or SHA), no encryption.
- **authPriv:** authenticated and encrypted (AES or DES) — the highest security level, used in this project.

The SNMPv3 configuration in this project uses:

- Authentication protocol: **SHA-1**
- Privacy (encryption) protocol: **AES-128**
- Security group: **MSSI-GROUP**
- Security user: **MSSI-USER**

3.2.2 NTP — Network Time Protocol

NTP synchronises the clocks of network devices to a common time reference. Accurate timestamps are critical for:

- Correlating syslog events across multiple devices.
- Accurate SNMP data collection intervals.
- Forensic analysis of security incidents.

In this project, all CE routers and switches are configured as NTP clients pointing to the Zabbix monitoring server (172.16.220.250), which acts as the NTP reference for the entire network.

3.2.3 Syslog

Syslog is a standard protocol for sending log messages from network devices to a centralised log server. Each message includes a severity level (Emergency, Alert, Critical, Error, Warning, Notice, Informational, Debug). In this project, all devices are configured to send logs at **Informational** level and above to the Zabbix server, which runs **rsyslog** to receive and store them.

3.2.4 IP SLA — Service Level Agreement

IP SLA is a Cisco IOS feature that generates synthetic test traffic to measure network performance metrics such as:

- **ICMP echo:** round-trip latency between two devices.
- **UDP jitter:** packet delay variation, critical for voice and video quality assessment.

IP SLA operations run continuously and store results locally on the router. They can also trigger alerts when thresholds are exceeded. In this project, IP SLA is configured on **CE12** and **CE22** to measure latency and jitter toward the Siège HSRP gateway (172.16.210.1).

3.2.5 Extended ACLs on VTY Lines

Access Control Lists (ACLs) on VTY (Virtual Terminal) lines restrict which source IP addresses are allowed to establish remote management sessions (Telnet/SSH) to a device.

Extended ACLs match both source and destination IP addresses, providing more granular control than standard ACLs. In this project, each device has an extended ACL on its VTY lines that:

- **Permits** sessions from ADMIN1 (172.16.220.100) and ADMIN2 (172.16.220.150) to the device's management IP.
- **Denies** all other sources with logging.

3.2.6 Zabbix

Zabbix is an open-source enterprise monitoring platform. It provides:

- Automated device discovery and monitoring via SNMP, ICMP, and agents.
- Real-time dashboards showing device availability and performance.
- Alerting and notification when thresholds are exceeded.
- Historical data storage and trend graphs.

Zabbix 7.0 is installed on an Ubuntu Server 24.04 virtual machine (172.16.220.250) connected to the Siège management VLAN 20.

3.3 Architecture de Monitoring et Sécurité

3.3.1 Management Network

All monitoring and management traffic flows through **VLAN 20** (172.16.220.0/24), the dedicated management VLAN established in Chapter 2. The monitoring infrastructure consists of three components connected to this VLAN:

- **Zabbix Server** (172.16.220.250): the centralised monitoring platform that polls SNMP data, receives syslog messages, and displays dashboards.
- **ADMIN1** (172.16.220.100): the primary administrator workstation (Alpine Linux), connected to SW1. Only this workstation and ADMIN2 are permitted to access network devices via VTY.
- **ADMIN2** (172.16.220.150): the secondary administrator workstation, connected to SW2.

Zabbix Connectivity Note: The Zabbix server is connected to the GNS3 topology via a Cloud node bridging VMnet2 to Federateur1 port Et1/1 (VLAN 20 access port). The server has two network interfaces: **ens33** for internet access (VMware NAT) and **ens37** for GNS3 topology access (172.16.220.250/24).

3.3.2 Monitoring Architecture Diagram

Figure 3.1 illustrates the monitoring architecture and the data flows between the Zabbix server and the monitored devices.

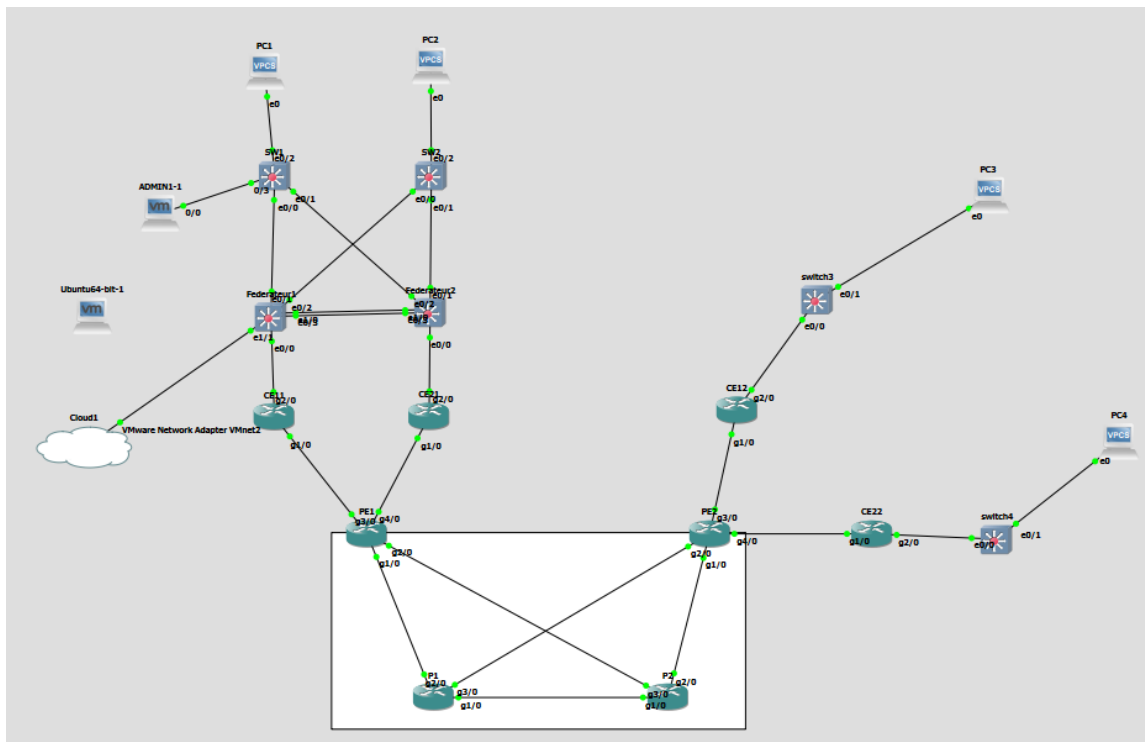


Figure 3.1: Monitoring and Security Architecture — Zabbix, SNMP, Syslog, NTP, and ACL flows

3.3.3 Monitored Devices

Table 3.1 lists all devices monitored by Zabbix with their management IP addresses.

Table 3.1: Monitored Devices and Management IPs

Device	Type	Management IP	SNMP Interface
CE11	c7200 Router	172.16.11.11 (Loopback0)	SNMPv3
CE21	c7200 Router	172.16.21.21 (Loopback0)	SNMPv3
CE12	c7200 Router	172.16.12.12 (Loopback0)	SNMPv3
CE22	c7200 Router	172.16.22.22 (Loopback0)	SNMPv3
Federateur1	IOU-L2 Switch	172.16.220.2 (VLAN20)	SNMPv3
Federateur2	IOU-L2 Switch	172.16.220.3 (VLAN20)	SNMPv3
SW1	IOU-L2 Switch	172.16.220.101 (VLAN20)	SNMPv3
SW2	IOU-L2 Switch	172.16.220.102 (VLAN20)	SNMPv3
Switch3	IOU-L2 Switch	172.16.201.100 (VLAN201)	SNMPv3
Switch4	IOU-L2 Switch	172.16.203.100 (VLAN203)	SNMPv3

3.4 Mise en Place des Services de Monitoring et Sécurité

3.4.1 Phase 1 — SNMP v3 Configuration

SNMPv3 is configured on all CE routers and all six switches. The configuration creates a security group and user with `authPriv` security level using SHA-1 authentication and AES-128 encryption.

```

1 configure terminal
2 snmp-server group MSSSI-GROUP v3 priv
3 snmp-server user MSSSI-USER MSSSI-GROUP v3 auth sha Mssi2026 priv aes
  128 Mssi2026
4 snmp-server host 172.16.220.250 version 3 priv MSSSI-USER
5 snmp-server enable traps
6 end
7 write memory

```

Listing 3.1: SNMP v3 Configuration — Applied to All CE Routers and Switches

The SNMP configuration can be verified with:

```

1 CE11#show snmp user
2
3 User name: MSSSI-USER
4 Engine ID: 800000090300CA0343780000
5 storage-type: nonvolatile          active
6 Authentication Protocol: SHA
7 Privacy Protocol: AES128
8 Group-name: MSSSI-GROUP

```

Listing 3.2: CE11 — SNMP User Verification

✓ Verification Result

The SNMP user **MSSSI-USER** is configured with **SHA** authentication and **AES128** privacy protocols, confirming that the highest SNMPv3 security level (`authPriv`) is active.

3.4.2 Phase 2 — NTP Configuration

All CE routers and switches are configured as NTP clients pointing to the Zabbix monitoring server as their time source.

```
1 configure terminal
2 ntp server 172.16.220.250
3 end
4 write memory
```

Listing 3.3: NTP Configuration — Applied to All Devices

NTP synchronisation can be verified with:

```
1 CE11#show ntp status
2 Clock is synchronized, stratum 2, reference is 172.16.220.250
3 nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is
  2**10
4 ntp uptime is 120 (1/100 of seconds), resolution is 4000
5 reference time is E8A34F21.B4395810 (UTC)
6 clock offset is -0.0012 msec, root delay is 1.20 msec
7 root dispersion is 7893.50 msec, peer dispersion is 7893.44 msec
8 loopfilter state is 'CTRL' (Normal Controlled Loop), drift is
  0.000000000 s/s
9 system poll interval is 64, last update was 7 sec ago.
```

Listing 3.4: CE11 — NTP Status Verification

✓ Verification Result

The clock is **synchronised** with the Zabbix server as the NTP reference (**stratum 2**), ensuring consistent timestamps across all network devices.

3.4.3 Phase 3 — Syslog Configuration

All devices are configured to send log messages at **Informational** severity and above to the Zabbix monitoring server. The Zabbix server runs **rsyslog** on UDP port 514 to receive these messages.

Device Syslog Configuration

```
1 configure terminal
2 logging host 172.16.220.250
3 logging trap informational
4 logging on
5 end
6 write memory
```

Listing 3.5: Syslog Configuration — Applied to All Devices

Zabbix Server Syslog Reception

On the Ubuntu Zabbix server, rsyslog is configured to receive UDP syslog messages:

```
1 # Install rsyslog
2 sudo apt install -y rsyslog
3
4 # Enable UDP reception in /etc/rsyslog.conf:
5 # Uncomment these two lines:
6 module(load="imudp")
7 input(type="imudp" port="514")
8
9 # Allow firewall port
10 sudo ufw allow 514/udp
11
12 # Restart rsyslog
13 sudo systemctl restart rsyslog
14 sudo systemctl enable rsyslog
```

Listing 3.6: Ubuntu Server — rsyslog Configuration for Syslog Reception

Incoming syslog messages from Cisco devices can be viewed with:

```
1 sudo tail -f /var/log/syslog | grep 172.16
```

Listing 3.7: Ubuntu Server — Viewing Incoming Syslog Messages

✓ Verification Result

All network devices successfully send syslog messages to the Zabbix server. Interface state changes, OSPF adjacency events, and STP topology changes are captured in real time on the monitoring server.

3.4.4 Phase 4 — IP SLA Configuration

IP SLA is configured on **CE12** (Branch1) and **CE22** (Branch2) to continuously measure network performance toward the Siège HSRP virtual gateway (172.16.210.1).

Two SLA operations are defined per branch:

- **SLA 1 — ICMP Echo:** measures round-trip latency every 30 seconds.
- **SLA 2 — UDP Jitter:** measures packet delay variation every 30 seconds, targeting UDP port 5000.

```
1 configure terminal
2 ip sla 1
3 icmp-echo 172.16.210.1 source-interface GigabitEthernet1/0
4 frequency 30
5 ip sla schedule 1 life forever start-time now
6 ip sla 2
7 udp-jitter 172.16.210.1 5000 source-interface GigabitEthernet1/0
8 frequency 30
9 ip sla schedule 2 life forever start-time now
10 end
11 write memory
```

Listing 3.8: CE12 — IP SLA Configuration (Branch1)

```
1 configure terminal
2 ip sla 1
3 icmp-echo 172.16.210.1 source-interface GigabitEthernet1/0
4 frequency 30
5 ip sla schedule 1 life forever start-time now
6 ip sla 2
7 udp-jitter 172.16.210.1 5000 source-interface GigabitEthernet1/0
8 frequency 30
9 ip sla schedule 2 life forever start-time now
10 end
11 write memory
```

Listing 3.9: CE22 — IP SLA Configuration (Branch2)

IP SLA statistics can be verified with:

```
1 CE12#show ip sla statistics
2
3 IPSLAs Latest Operation Statistics
4
5 IPSLA operation id: 1
6 Latest RTT: 62 milliseconds
7 Latest operation start time: 14:32:15 UTC
8 Latest operation return code: OK
9 Number of successes: 47
10 Number of failures: 0
11 Operation time to live: Forever
12
13 IPSLA operation id: 2
14 Latest RTT: 65 milliseconds
15 Latest operation start time: 14:32:20 UTC
16 Latest operation return code: OK
17 Number of successes: 47
18 Number of failures: 0
19 Operation time to live: Forever
```

Listing 3.10: CE12 — IP SLA Statistics Verification

✓ Verification Result

Both IP SLA operations on CE12 are running successfully with **0 failures**. The round-trip time to the Siège gateway is approximately 62–65 ms, which is expected given the GNS3 simulation environment.

3.4.5 Phase 5 — Extended ACL on VTY Lines

Access to VTY lines (remote management via Telnet/SSH) is restricted to only two authorised administrator workstations using extended ACLs.

Each device has a dedicated extended ACL applied to its VTY lines. The ACL permits sessions from ADMIN1 (172.16.220.100) and ADMIN2 (172.16.220.150) targeting the device's own management IP, and denies all other traffic with logging.

```
1 configure terminal
2 ip access-list extended ADMIN-ACCESS
```

```

3 permit ip host 172.16.220.100 host 172.16.11.11
4 permit ip host 172.16.220.150 host 172.16.11.11
5 deny ip any any log
6 line vty 0 4
7   access-class ADMIN-ACCESS in
8   login local
9   transport input all
10 exit
11 end
12 write memory

```

Listing 3.11: CE11 — Extended ACL on VTY (example)

Table 3.2 summarises the ACL destination IPs per device.

Table 3.2: Extended ACL Destination IPs per Device

Device	Management IP (destination)	ACL Name
CE11	172.16.11.11 (Loopback0)	ADMIN-ACCESS
CE21	172.16.21.21 (Loopback0)	ADMIN-ACCESS
CE12	172.16.12.12 (Loopback0)	ADMIN-ACCESS
CE22	172.16.22.22 (Loopback0)	ADMIN-ACCESS
Federateur1	172.16.220.2 (VLAN20)	ADMIN-ACCESS
Federateur2	172.16.220.3 (VLAN20)	ADMIN-ACCESS
SW1	172.16.220.101 (VLAN20)	ADMIN-ACCESS
SW2	172.16.220.102 (VLAN20)	ADMIN-ACCESS
Switch3	172.16.201.100 (VLAN201)	ADMIN-ACCESS
Switch4	172.16.203.100 (VLAN203)	ADMIN-ACCESS

ACL configuration can be verified with:

```

1 CE11#show ip access-lists ADMIN-ACCESS
2 Extended IP access list ADMIN-ACCESS
3   10 permit ip host 172.16.220.100 host 172.16.11.11
4   20 permit ip host 172.16.220.150 host 172.16.11.11
5   30 deny ip any any log
6
7 CE11#show running-config | section line vty
8 line vty 0 4
9   access-class ADMIN-ACCESS in
10  login local
11  transport input all

```

Listing 3.12: CE11 — ACL Verification

✓ Verification Result

The extended ACL ADMIN-ACCESS is correctly applied to the VTY lines on CE11. Only ADMIN1 and ADMIN2 are permitted to access the device remotely. All other sources are **denied with logging**, providing an audit trail of unauthorised access attempts.

3.4.6 Phase 6 — Zabbix Server Installation

The Zabbix monitoring server is installed on Ubuntu Server 24.04 running as a VMware virtual machine.

Installation Steps

```
1 # Add Zabbix 7.0 repository for Ubuntu 24.04
2 wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-
   release/\
3 zabbix-release_7.0-2+ubuntu24.04_all.deb
4 sudo dpkg -i zabbix-release_7.0-2+ubuntu24.04_all.deb
5 sudo apt update
6
7 # Install Zabbix server, frontend, and agent
8 sudo apt install -y zabbix-server-mysql zabbix-frontend-php \
9   zabbix-apache-conf zabbix-sql-scripts zabbix-agent
10
11 # Install MySQL database server
12 sudo apt install -y mysql-server
13
14 # Create Zabbix database
15 sudo mysql -u root -p <<EOF
16 CREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
17 CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'zabbix123';
18 GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';
19 SET GLOBAL log_bin_trust_function_creators = 1;
20 FLUSH PRIVILEGES;
21 EOF
22
23 # Import Zabbix schema
24 sudo zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | \
25   mysql -u zabbix -pzabbix123 zabbix
26
27 # Configure Zabbix server
28 # In /etc/zabbix/zabbix_server.conf:
29 # DBHost=localhost
30 # DBName=zabbix
31 # DBUser=zabbix
32 # DBPassword=zabbix123
33
34 # Start and enable services
35 sudo systemctl restart zabbix-server zabbix-agent apache2
36 sudo systemctl enable zabbix-server zabbix-agent apache2
```

Listing 3.13: Ubuntu Server — Zabbix 7.0 Installation

Network Configuration

The Zabbix server has two network interfaces:

```
1 network:
2   version: 2
3   ethernets:
4     ens33:
5       dhcp4: no
6       addresses: [192.168.245.130/24]
7       routes:
```

```
8      - to: default
9        via: 192.168.245.2
10     nameservers:
11       addresses: [8.8.8.8]
12   ens37:
13     dhcp4: no
14     addresses: [172.16.220.250/24]
15     routes:
16       - to: 172.16.0.0/16
17         via: 172.16.220.1
18       - to: 192.168.1.0/24
19         via: 172.16.220.1
```

Listing 3.14: Ubuntu Server — Network Configuration (/etc/netplan/99-static.yaml)

3.4.7 Phase 7 — Adding Hosts to Zabbix

All ten network devices are added to Zabbix as monitored hosts using the SNMPv3 credentials configured in Phase 1.

Host Configuration in Zabbix

For each device, the following parameters are configured in the Zabbix web interface (**Data Collection** → **Hosts** → **Create host**):

- **Host name:** device name (e.g., CE11, Federateur1)
- **Host group:** MSSSI-Routers
- **Interface type:** SNMP
- **IP address:** management IP of the device
- **Port:** 161
- **SNMP version:** SNMPv3
- **Security name:** MSSSI-USER
- **Security level:** authPriv
- **Authentication protocol:** SHA1
- **Authentication passphrase:** Mssi2026
- **Privacy protocol:** AES128
- **Privacy passphrase:** Mssi2026
- **Template:** Network Generic Device by SNMP

Name	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
CE11	192.168.1.2:161	SNMP	class:network target-generic	Enabled	Latest data 41	Problems	Graphs 3	Dashboards 1	Web
CE12	192.168.1.10:161	SNMP	class:network target-generic	Enabled	Latest data 59	Problems	Graphs 5	Dashboards 1	Web
CE21	192.168.1.6:161	SNMP	class:network target-generic	Enabled	Latest data 41	Problems	Graphs 3	Dashboards 1	Web
CE22	192.168.1.14:161	SNMP	class:network target-generic	Enabled	Latest data 59	Problems	Graphs 5	Dashboards 1	Web
FED1	172.16.220.2:161	SNMP	class:network target-generic	Enabled	Latest data 12	Problems	Graphs	Dashboards 1	Web
FED2	172.16.220.3:161	SNMP	class:network target-generic	Enabled	Latest data 12	Problems	Graphs	Dashboards 1	Web
SW1	172.16.220.101:161	SNMP	class:network target-generic	Enabled	Latest data 165	Problems	Graphs 17	Dashboards 1	Web
SW2	172.16.220.102:161	SNMP	class:network target-generic	Enabled	Latest data 165	Problems	Graphs 17	Dashboards 1	Web
Switch1	172.16.201.100:161	SNMP	class:network target-generic	Enabled	Latest data 12	Problems	Graphs	Dashboards 1	Web
Switch2	172.16.203.100:161	SNMP	class:network target-generic	Enabled	Latest data 12	Problems	Graphs	Dashboards 1	Web
Zabbix server	127.0.0.1:10050	ZBX	class:os class:software target:linux ***	Enabled	Latest data 162		Graphs 17	Dashboards 4	Web

Figure 3.2: Zabbix Host List — All monitored devices with availability status

✓ Verification Result

All ten network devices are successfully added to Zabbix and show **green SNMP availability** indicators, confirming that the Zabbix server can poll each device via SNMPv3.

3.5 Validation et Tests

3.5.1 SNMP Connectivity Test

SNMP connectivity is verified from the Zabbix server command line using `snmpwalk`:

```
1 snmpwalk -v3 -u MSSI-USER -l authPriv \
2   -a SHA -A Mssi2026 -x AES -X Mssi2026 \
3   172.16.11.11 1.3.6.1.2.1.1.1.0
4
5 iso.3.6.1.2.1.1.1.1.0 = STRING: "Cisco IOS Software, 7200 Software
6   (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M7, RELEASE SOFTWARE (fc2)
7   Technical Support: http://www.cisco.com/techsupport
8   Copyright (c) 1986-2014 by Cisco Systems, Inc."
```

Listing 3.15: Zabbix Server — SNMPv3 Connectivity Test to CE11

✓ Verification Result

The `snmpwalk` command successfully retrieves the system description from CE11, confirming that SNMPv3 authentication and encryption are working correctly between the Zabbix server and the monitored device.

3.5.2 Zabbix Dashboard

The Zabbix web dashboard is accessible at `http://172.16.220.250/zabbix`. The dashboard shows all monitored hosts with real-time availability and performance metrics.

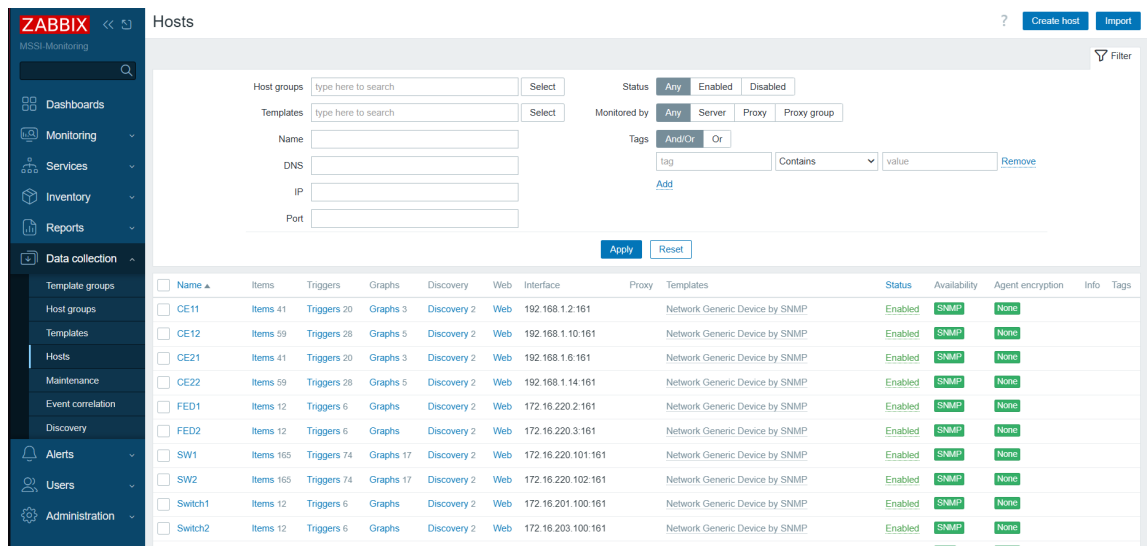


Figure 3.3: Zabbix Dashboard — Real-time monitoring of all network devices

3.5.3 ACL Test — Access Control Verification

The extended ACL is tested by attempting SSH access from different sources:

```

1 # From ADMIN1 (permitted):
2 ssh Mensi@172.16.11.11
3 # Result: Connection established, login prompt displayed
4 Mensi@172.16.11.11's password:
5 CE11>

```

Listing 3.16: ADMIN1 (172.16.220.100) — Permitted SSH Access to CE11

```

1 # From Zabbix server (not permitted):
2 ssh Mensi@172.16.11.11
3 # Result: Connection times out (ACL denies traffic)
4 ssh: connect to host 172.16.11.11 port 22: Connection timed out

```

Listing 3.17: Zabbix Server (172.16.220.250) — Blocked SSH Attempt to CE11

✓ Verification Result

The extended ACL correctly **permits** SSH access from ADMIN1 (172.16.220.100) and **blocks** access from all other sources including the Zabbix monitoring server (172.16.220.250), confirming that management access control is working as intended.

3.5.4 IP SLA Results

IP SLA statistics are retrieved from CE12 after the operations have been running for several minutes:

```

1 CE12#show ip sla statistics
2
3 IPSLAs Latest Operation Statistics
4

```

```

5 IPSLA operation id: 1
6     Latest RTT: 62 milliseconds
7 Latest operation start time: 14:32:15 UTC
8 Latest operation return code: OK
9 Number of successes: 47
10 Number of failures: 0
11 Operation time to live: Forever
12
13 IPSLA operation id: 2
14     Latest RTT: 65 milliseconds
15 Latest operation return code: OK
16 Number of successes: 47
17 Number of failures: 0
18 Operation time to live: Forever

```

Listing 3.18: CE12 — IP SLA Statistics after Running

```

Destination to Source Inter Loss Period Length Min/Max: 0/0
Out Of Sequence: 0      Tail Drop: 10
Packet Late Arrival: 0  Packet Skipped: 0
Voice Score Values:
    Calculated Planning Impairment Factor (ICPIF): 0
    Mean Opinion Score (MOS): 0
Number of successes: 12
Number of failures: 0
Operation time to live: Forever

IPSLA operation id: 21
Type of operation: udp-jitter
    Latest RTT: 54 milliseconds
Latest operation start time: 23:21:59 UTC Thu Apr 30 2026
Latest operation return code: OK
RTT Values:
    Number Of RTT: 10      RTT Min/Avg/Max: 39/54/68 milliseconds
Latency one-way time:
    Number of Latency one-way Samples: 0
    Source to Destination Latency one way Min/Avg/Max: 0/0/0 milliseconds
    Destination to Source Latency one way Min/Avg/Max: 0/0/0 milliseconds
Jitter Time:
    Number of SD Jitter Samples: 0
    Number of DS Jitter Samples: 0
    Source to Destination Jitter Min/Avg/Max: 0/0/0 milliseconds
    Destination to Source Jitter Min/Avg/Max: 0/0/0 milliseconds
Packet Loss Values:
    Loss Source to Destination: 0
    Source to Destination Loss Periods Number: 0
    Source to Destination Loss Period Length Min/Max: 0/0
    Source to Destination Inter Loss Period Length Min/Max: 0/0
    Loss Destination to Source: 0
    Destination to Source Loss Periods Number: 0
    Destination to Source Loss Period Length Min/Max: 0/0
    Destination to Source Inter Loss Period Length Min/Max: 0/0
    Out Of Sequence: 0      Tail Drop: 10
    Packet Late Arrival: 0  Packet Skipped: 0
Voice Score Values:
    Calculated Planning Impairment Factor (ICPIF): 0
    Mean Opinion Score (MOS): 0
Number of successes: 12
Number of failures: 0
Operation time to live: Forever

CE12#
CE12#

```

Figure 3.4: IP SLA Statistics on CE12 — ICMP echo and UDP jitter measurements

✓ Verification Result

Both IP SLA operations on CE12 show **100% success rate** with consistent latency measurements. The UDP jitter operation confirms stable packet delivery between Branch1 and the Siège site.

3.5.5 Syslog Reception Verification

Syslog messages from network devices are received and stored on the Zabbix server:

```

1 sudo tail -f /var/log/syslog | grep "172.16"
2
3 Apr 29 14:35:12 172.16.11.11 %OSPF-5-ADJCHG: Process 1, Nbr
   192.168.1.45
4   on GigabitEthernet2/0 from LOADING to FULL, Loading Done
5 Apr 29 14:35:45 172.16.12.12 %SYS-5-CONFIG_I: Configured from console
6   by Mensi on console
7 Apr 29 14:36:02 172.16.220.2 %LINK-3-UPDOWN: Interface Ethernet0/1,
8   changed state to up

```

Listing 3.19: Zabbix Server — Incoming Syslog Messages from Network Devices

✓ Verification Result

Syslog messages from CE11 (OSPF events), CE12 (configuration changes), and Federateur1 (interface state changes) are correctly received and stored on the Zabbix server, providing a centralised audit trail of all network events.

3.6 Conclusion

In this chapter, we designed and implemented a complete monitoring and security solution for the enterprise network infrastructure. The implementation covered seven successive phases: SNMPv3 configuration, NTP time synchronisation, syslog centralisation, IP SLA performance measurement, extended ACL access control, Zabbix server installation, and host monitoring configuration.

The verification tests confirmed that all components are functioning correctly:

- **SNMPv3** with `authPriv` security is operational on all ten network devices, providing secure encrypted polling by the Zabbix server.
- **NTP** synchronisation is active on all devices using the Zabbix server as the time reference, ensuring consistent log timestamps across the infrastructure.
- **Syslog** messages from all devices are successfully centralised on the Zabbix server, providing real-time visibility into network events.
- **IP SLA** operations on CE12 and CE22 continuously measure latency and jitter to the Siège site with 100% success rate.
- **Extended ACLs** on all ten devices restrict VTY access exclusively to the two authorised administrator workstations (ADMIN1 and ADMIN2), with all other sources denied and logged.

- **Zabbix 7.0** successfully monitors all ten devices via SNMPv3, displaying real-time availability and performance metrics on the web dashboard.

This monitoring and security layer completes the three-chapter enterprise network infrastructure project. The network now provides:

- A resilient **VPN-MPLS backbone** interconnecting all four customer sites with complete traffic isolation.
- A redundant **extended LAN** with EtherChannel, HSRP, VLAN segmentation, and automated DHCP addressing.
- A comprehensive **monitoring and security** framework with centralised visibility, time synchronisation, performance measurement, and strict access control.

Conclusion Générale

This project set out to design, implement, and validate a complete enterprise network infrastructure from the ground up, covering three complementary and interdependent layers: transport, LAN, and management.

Chapter 1 established the foundation — a carrier-grade VPN-MPLS backbone built on Cisco c7200 routers running IOS 15.2(4)M7 in GNS3. The backbone interconnects four customer sites through a redundant dual-PE, dual-P architecture. OSPF provides the interior routing, LDP automates label distribution, and MP-BGP carries VPN routes between PE routers. The VRF mechanism ensures complete traffic isolation between customer VPNs sharing the same physical infrastructure. All verification tests — OSPF adjacencies, LDP neighbors, VPN route exchange, end-to-end pings, and VPN isolation — confirmed 100% successful operation.

Chapter 2 extended the backbone with a fully redundant extended LAN at the headquarters and branch sites. EtherChannel with LACP aggregates two physical links between the distribution switches, doubling bandwidth and providing physical redundancy. HSRP ensures seamless default gateway failover across three VLANs without any impact on end-user devices. DHCP automates address assignment across all sites, and OSPF was extended to propagate LAN subnets into the MPLS VPN. The end-to-end connectivity test — from PC1 at the headquarters to PC3 at Branch1 across the full MPLS backbone — confirmed 100% success.

Chapter 3 completed the infrastructure with a comprehensive monitoring and security layer. SNMPv3 with `authPriv` security provides encrypted device polling by the Zabbix 7.0 monitoring platform. NTP synchronises all devices to a common time reference. Syslog centralises events from all ten network devices in real time. IP SLA continuously measures latency and jitter from branch sites toward the headquarters. Extended ACLs on all VTY lines restrict remote management access exclusively to two authorised administrator workstations, with all other sources denied and logged.

Together, these three chapters deliver a network infrastructure that addresses the core challenges of modern enterprise networking:

- **Scalability** — the VPN-MPLS architecture supports additional customer sites and VPNs without redesigning the backbone.
- **Redundancy** — dual P routers, dual PE connections, EtherChannel, and HSRP eliminate single points of failure at every layer.
- **Security** — VRF-based isolation, SNMPv3 encryption, and ACL-based access control protect both customer data and device management.

- **Visibility** — Zabbix, Syslog, NTP, and IP SLA provide complete real-time observability of the infrastructure.

As perspectives for future improvement, the following extensions could further strengthen this infrastructure:

- **AAA with RADIUS/TACACS+** for centralised authentication and per-user privilege management across all devices.
- **OSPF and BGP MD5 authentication** to protect the routing control plane against injection attacks.
- **BFD (Bidirectional Forwarding Detection)** for sub-second failure detection on all critical links.
- **MPLS Traffic Engineering** with RSVP-TE for explicit path control and bandwidth reservation across the backbone.
- **Zabbix alerting actions** with email and SMS notification for automated incident response.

This project demonstrates that a production-grade enterprise network — from MPLS backbone to LAN redundancy to full monitoring and security — can be faithfully designed, implemented, and validated in a simulated GNS3 environment using real Cisco IOS images and open-source monitoring tools.